SparkShield at ArA-DF 2026: Robust Arabic Speech Deepfake Detection via Realistic Audio Augmentation and Multi-Scale XLS-R Score Fusion

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Abstract

This paper presents the SparkShield system submitted to ArA-DF 2026 Track 2, which focuses on robust Arabic speech deepfake detection under realistic acoustic degradations. To improve robustness, we apply a comprehensive augmentation pipeline involving background noise, reverberation, codec compression, re-encoding, and volume perturbation. Our system consists of three parallel subsystems using XLS-R models with 300M, 1B, and 2B parameters as front-end feature extractors. Each subsystem employs learnable layer-wise feature fusion, followed by a Multi-kernel gated Convolution (MultiConv) module and Multi-Head Attention Pooling (MHAP) for utterance-level spoof detection. The output logits of the three independently trained subsystems are finally fused. The proposed system achieves an EER of 0.6575% on the evaluation set and ranks first in ArA-DF 2026 Track 2.

1 Introduction

Recent advances in neural speech generation technologies have significantly improved the quality and naturalness of synthetic speech. Modern text-to-speech (TTS) and voice conversion (VC) systems are capable of producing highly realistic speech that closely resembles human recordings (Du et al., 2025; Chen et al., 2026; Liu, 2024; Zheng et al., 2026). Although these technologies provide valuable applications in human-computer interaction and personalized speech services, they also introduce potential security risks, including speaker impersonation, voice fraud, and the generation of misleading audio content. Therefore, reliable speech deepfake detection has become an important research problem in speech processing and multimedia security.

Many speech deepfake detection studies are conducted under relatively controlled acoustic conditions. However, detecting deepfake speech in real-world environments remains challenging because background noise, reverberation, codec compression, bandwidth limitations, and social-media re-encoding can obscure the artifacts introduced by speech generation systems. Such degradations may suppress subtle synthesis traces and cause a substantial mismatch between clean training data and real-world audio. Recent benchmark efforts, such as ASVspoof 2021, ASVspoof 5 and Audio Deepfake Detection (ADD) challenge, have started to consider more challenging scenarios involving environmental variations and transmission effects (Yamagishi et al., 2021; Wang et al., 2024; Yi et al., 2022, 2023). Nevertheless, existing evaluations have focused primarily on English speech, leaving other languages underexplored. Arabic speech introduces additional challenges because of its dialectal diversity and distinctive phonetic characteristics. Accordingly, ArA-DF 2026 Track 2 evaluates detection robustness under realistic acoustic distortions, providing a practical benchmark for real-world deployment.

In this paper, we present the SparkShield system for ArA-DF 2026 Track 2. Our approach combines realistic audio augmentation with complementary representations extracted from XLS-R models of different scales (Babu et al., 2022). Each subsystem applies learnable layer-wise feature fusion and a MultiConv–MHAP classifier (Tran et al., 2025) to capture spoofing artifacts over different temporal contexts. Finally, the outputs of the three independently trained subsystems are fused to improve detection robustness. Experimental results demonstrate the effectiveness of the proposed approach under realistic acoustic degradations.

2 Task Description and Dataset

2.1 Task Description

We participate in Track 2 of the ArA-DF 2026 shared task, the first benchmark dedicated to robust

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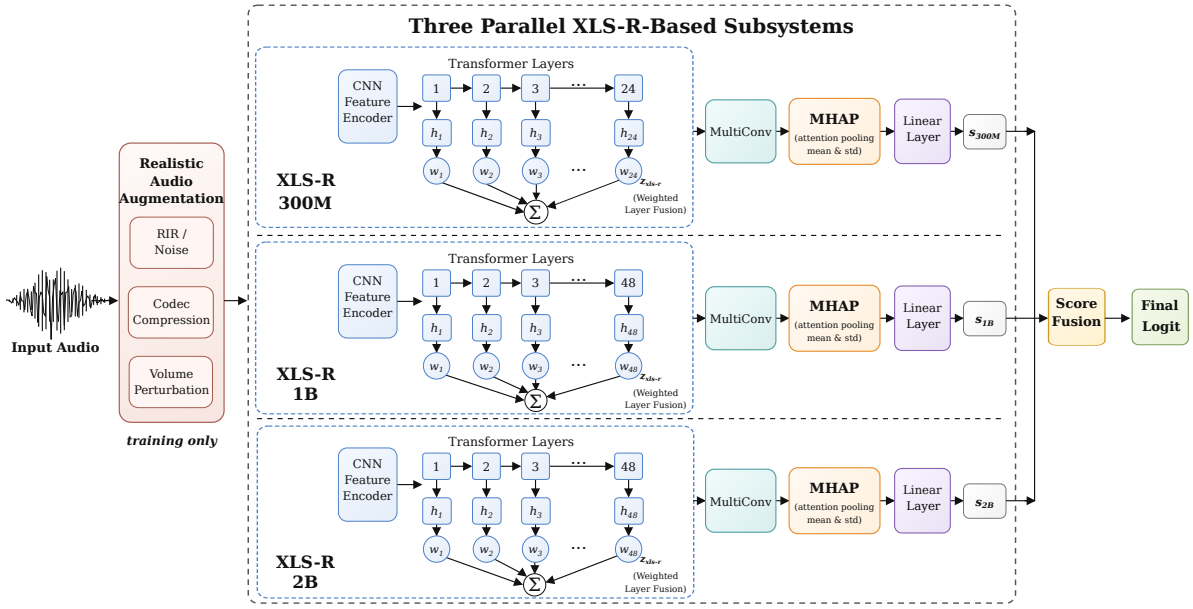


Figure 1: Overview of the proposed system. Realistic audio augmentation simulates environmental and transmission degradations during training. Each subsystem uses an XLS-R front end with learnable layer-wise weighted fusion, followed by a MultiConv–MHAP back end. The scores of the three subsystems are fused to produce the final logit.

Arabic audio deepfake detection. This track targets acoustic robustness: participating systems are evaluated on speech samples corrupted by realistic, real-world distortions, including codec compression, additive background noise, and social-media re-encoding. The task is formulated as a binary classification problem. Each input audio segment is classified as either spoofed or bonafide:

- **Spoofed speech**: synthetic speech from TTS or VC.
- **Bonafide speech**: genuine human speech from native Arabic speakers.

Participants are required to submit logit scores corresponding to the bonafide class, where higher scores indicate a greater probability that the speech is genuine. The primary evaluation metric is the Equal Error Rate (EER), with lower EER representing better detection performance under degraded acoustic conditions.

2.2 Dataset

The dataset is divided into train, dev, dev-test, and test sets containing 22,500, 21,000, 14,193, and 127,746 segments, respectively. The train and dev sets provide labels and dialect-family and gender information in their filenames, while the two test sets are unlabeled and anonymized. All recordings are 16-kHz mono PCM audio. Overall, the mean and median durations across all splits fall within the range of 9.0-9.6 seconds, showing consistent central tendency across splits. Most utterances are between 5 and 15s long. Apart from duration statistics, we also analyze the quantity distribution of bonafide and spoofed audio samples, where bonafide-to-spoof ratio is approximately 1:2. These observations provide practical guidelines for model training and inference.

3 Systems Overview

Figure 1 provides an overview of the proposed system. The system consists of three main components: realistic audio augmentation, XLS-R-based feature extraction with layer-wise weighted fusion, and multi-system score fusion.

3.1 Data Augmentation

We employ a hierarchical data augmentation strategy during training to enhance model robustness. The augmentation pipeline consists of three independent layers: (1) acoustic environment augmentation (80% probability) applying either room impulse response (Ko et al., 2017) or background noise (SNR: 1-20 dB) (Snyder et al., 2015); (2) codec compression (80% probability) using either MP3 compression (32-128 kbps) or random codec chains with multiple encoding passes; and (3) volume adjustment (30% probability) with gain ranging from -2.5 to +2.0 dB, corresponding to ampli-

tude scaling factors of approximately 0.75 to 1.26. This approach simulates real-world audio degradation from both the recording environment (e.g., room impulse response and background noise) and the transmission chain (e.g., codec compression and cross-platform re-encoding).

3.2 Model Architecture

3.2.1 Front-end Feature Extractor

We adopt pre-trained XLS-R (Babu et al., 2022) as our front-end speech feature extractor. XLS-R is a cross-lingual self-supervised pre-trained model built on the wav2vec 2.0 paradigm, pre-trained on over 436k hours of speech covering 128 languages, which delivers robust generalizable representations for low-resource Arabic speech under degraded acoustic conditions.

To enrich feature diversity and strengthen anti-degradation capability, we employ three differently sized XLS-R variants with 300M, 1B and 2B parameters separately. Each variant takes raw 16 kHz mono waveforms as input and outputs frame-level contextual speech embeddings from multiple Transformer layers. Instead of using only the final-layer representation, we introduce a learnable layer-wise weighted fusion strategy to adaptively aggregate the hidden representations from different layers. Specifically, the outputs of all Transformer layers are weighted and summed to obtain a more discriminative fused representation for subsequent spoof detection. Within each subsystem, the XLS-R backbone and its downstream classifier are jointly fine-tuned, while the three subsystems are trained independently.

3.2.2 Back-end Classifier

We adopt an identical back-end classifier for each XLS-R branch, composed of MultiConv and MHAP (Tran et al., 2025). The MultiConv module utilizes convolution kernels of diverse sizes to simultaneously capture local transient artifacts and long-range contextual patterns introduced by speech synthesis and voice conversion. Gated convolution further suppresses irrelevant acoustic information and highlights discriminative spoofing clues within frame-level embeddings. Subsequently, MHAP aggregates sequential frame embeddings into a single fixed-dimensional utterance-level representation. Compared with simple average pooling, attention pooling adaptively assigns higher weights to frames containing critical fake-speech traces. Finally, a linear projection layer outputs the detection score. During inference, we obtain the final score by taking the unweighted average of their predicted bonafide logits.

4 Experiment Setup

All models are trained on two NVIDIA H20 GPUs. We adopt batch sizes of 48, 32, and 32 for the 300M, 1B, and 2B XLS-R variants, respectively. Models are trained using randomly sampled 10-second audio segments; shorter utterances are zero-padded to reach 10 seconds. During inference, we adopt a sliding window strategy with a window length of 18 seconds. For long utterances, five windows are extracted and the final prediction score is averaged; for audio shorter than 18 seconds, only a single window is used.

We adopt the AdamW (Loshchilov and Hutter, 2019) optimizer with discriminative module-wise learning rates. Model parameters are split into two modules: the front-end SSL feature extractor and the back-end classifier. We set separate maximum learning rates and weight decay coefficients for each module. Specifically, the front-end extractor is optimized with a maximum learning rate of $1e{-}5$ and weight decay of 0.001, while the classifier uses a higher maximum learning rate of $1e{-}4$ together with weight decay of 0.01. A OneCycle learning rate policy is employed to schedule the learning rate. The warm-up phase occupies the first 30% of total training iterations, where the learning rate linearly rises from $max_lr/10$ to the predefined peak value. Afterward, the learning rate gradually decays until reaching $max_lr/10000$ at the end of training.

We use a class-weighted cross-entropy loss to address label imbalance in the training set. Given the predominance of spoofed samples, class weights of 0.33 (spoof) and 0.67 (bonafide) are applied, assigning greater importance to the minority bonafide class.

5 Results and Analysis

5.1 Ablation Studies

We conduct ablation experiments to evaluate the effects of key model, augmentation, and inference choices, as summarized in **Table 1**.

The first group reproduces the official baseline (XLS-R-300M+AASIST). Our reproduction obtains an EER of 29.0294%, compared with 27.2097% for the official baseline, while adding RawBoost (Tak et al., 2022) and SpecAug

Table 1: Ablation studies of model, training configurations and inference strategies on *dev-test* of Track 2.

Model	Training Data	Augmentation	Inference Scheme	EER (%)
XLS-R-300M + AASIST	Train & 4s	None	Single 4s win	29.0294
	Train & 4s	Rawboost_algo5 + SpecAug	Single 4s win	28.6900
	Train & 10s	Rawboost_algo5 + SpecAug	Single 10s win	20.1763
XLS-R-300M + MultiConv + MHAP	Train & 10s	None	Single 10s win	6.9533
	Train+ArVoice & 10s	None	Single 10s win	15.6131
XLS-R-300M + MultiConv + MHAP	Train & 10s	RIR/Noise + Codec + Volume	Single 10s win	1.5038
			Avg of 5 × 10s win	1.2210
			Single 15s win	1.1361
			Avg of 5 × 15s win	1.1125
			Avg of 5 × 18s win	**0.9428**

Table 2: Performance comparison of different XLS-R-based systems on ArA-DF 2026 Track 2.

Model	EER (%)	
	dev-test	test
Official baseline	27.2097	27.1132
XLS-R-300M + MultiConv + MHAP	0.9428	0.9654
XLS-R-1B + MultiConv + MHAP	0.9947	1.0514
XLS-R-2B + MultiConv + MHAP	0.8014	0.8034
Fusion of Scores	**0.6553**	**0.6575**

(Park et al., 2019) yields only a marginal improvement. Extending both the training and inference window from 4s to 10s, however, reduces the EER substantially from 28.6900% to 20.1763%, confirming that longer speech segments provide more discriminative evidence for spoof detection.

Switching from the AASIST-based configuration (Jung et al., 2022) to MultiConv+MHAP reduces the EER from 20.1763% to 6.9533%, even without data augmentation, demonstrating the effectiveness of the adopted back end. Introducing the full augmentation pipeline described in Section 3.1 then drives the EER down sharply from 6.9533% to 1.5038%, underscoring the importance of simulating realistic acoustic degradations during training.

We further analyze the effect of inference window length and multi-window ensembling. Longer inference windows generally improve performance, while averaging predictions over multiple windows provides additional gains. The best configuration averages five 18-s windows and achieves an EER of 0.9428%.

Finally, we enrich the training set with publicly available Arabic synthetic speech, i.e., ArVoice (Toyin et al., 2025). Contrary to expectation, this leads to a clear degradation, which we attribute to the domain mismatch between the external synthetic data and the target evaluation distribution.

5.2 Results of Subsystems and Fused System

Table 2 summarizes the performance of our proposed systems on the development-phase dev-test set and evaluation-phase test set of ArA-DF 2026 Track 2. The official baseline yields EER values of 27.2097% and 27.1132% on these two partitions, respectively. Our three individual subsystems obtain EER values of 0.9428%, 0.9947% and 0.8014% on the dev-test set, and 0.9654%, 1.0514% and 0.8034% on the test set, respectively. Score fusion of the three models further reduces the EER to 0.6553% on dev-test, and achieves an EER of 0.6575% on the final evaluation test set. This result ranks our system first on the official leaderboard of the ArA-DF 2026 Track 2 challenge.

6 Conclusion

This paper presents our submission to the ArA-DF 2026 Track 2, which targets robust Arabic speech deepfake detection. To address the challenges, we utilize a comprehensive system built upon three key strategies: (1) extensive data augmentation simulating real-world acoustic conditions, (2) a parallel architecture consisting of three XLS-R-based subsystems, each equipped with a MultiConv+MHAP back-end classifier, and (3) multi-window inference that aggregates evidence from different regions of long utterances. Our ablation studies demonstrate that extending the training and inference window from 4s to 10s yields substantial performance gains, while the comprehensive augmentation pipeline significantly improves robustness by exposing the model to diverse degradation patterns during training. Fusion of predictions from three XLS-R-based subsystems further reduces the EER to 0.6553% on the dev-test set and 0.6575% on the final evaluation test set, securing first place in the challenge and verifying the effectiveness of our approach.

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